

MUHAMMAD IBRAHIM MUSYAFFA

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Fresh graduate in Applied Electronics Engineering from Universitas Negeri Yogyakarta with practical experience in industrial automation, electronics, robotics, embedded systems, IoT monitoring, and PCB design. Experienced in developing, assembling, installing, testing, and troubleshooting electronic systems, including machine condition monitoring and predictive maintenance solutions. Skilled in sensor integration, microcontroller programming, web-based monitoring systems, and applying Machine Learning for early fault detection.

Education

Universitas Negeri Yogyakarta - **Yogyakarta** September 2022 – September 2026
Bachelor of Applied Electronics Engineering

- GPA : 3.75/4.00
- Thesis : Design and Development of a Machine Learning-Based Predictive Maintenance System for Early Fault Detection in a Regular Shakeout Sub-Machine Motor.

Work Experience

PT Komatsu Indonesia - **Jakarta** July 2025 – October 2025
Internship – Foundry Maintenance
PT Komatsu Indonesia is a manufacturing company specializing in the production of heavy equipment for the construction and mining sectors.

- Developed and implemented an IoT-based Smart Lighting Control System for remote control and monitoring of lighting in the Foundry Plant area.
- Built a Spare Parts Order Status Monitoring website and established an ordering workflow based on operational needs and input from the spare parts manager.
- Assembled a Thermoelectric Generator (TEG) prototype circuit, calculated electrical parameters from input to output, conducted testing, and demonstrated the device.
- Developed a Machine Learning-based predictive maintenance system for the Mainshakeout Regular motor, monitoring vibration, temperature, sound, and electrical power through a web dashboard and abnormal-condition notification.
- Supported the Maintenance team's field activities, organized spare parts in the warehouse, and prepared training and assessment materials related to foundry machinery and equipment.

PT AM/NS Indonesia - **Bekasi** February 2025 – May 2025
Internship – Electrical Engineer
PT AM/NS Indonesia is a manufacturing company that processes cold-rolled steel, galvanized steel, and annealed steel in sheet and coil form.

- Analyzed thermocouple sensor functionality in the tube furnace reheating process on the Continuous Galvanizing Line (CGL) to support temperature control during galvanizing.
- Performed maintenance and repairs on electrical equipment, including repairing lights, repairing the steel sheet cutting machine, periodically checking panel voltage, assisting with motor drive inspections, assisting with motor lubrication, and other tasks to ensure operational reliability.

Organizational experience

UNY Robotics - **Yogyakarta** November 2023 – February 2025
Electronics and Pilot
UNY Robotics is a division of the Engineering Technology Student Club that focuses on technology, specifically robotics.

- Designing and developing the electronic systems required for robots, including component selection and assembly of electronic components.
- Testing and troubleshooting electronic circuits to ensure the robots operate without errors.
- Controlling and operating robots during competitions.

Electrical and Electronic Engineering Vocational Student Association (HMVE) UNY - **Yogyakarta** June 2023 - December 2023
Staff of the IPTEK Division
HMVE serves as a facilitator in developing academic and non-academic interests, channeling aspirations, and fostering a sense of social responsibility among Electrical and Electronics Engineering students.

- Created comprehensive materials offering practical and in-depth knowledge of electrical and electronic engineering, which were submitted to the Medinfo Division for posting on Instagram.
- Assisting in coordinating with travel agencies for the Division's programs, including tracking student participation in industry visits.
- Helped find speakers for workshops and served as a technical operator during the events, ensuring the smooth execution of programs related to the role of AI in industrial automation systems.
- Contribute to the Personnel Division's work program, particularly in introducing campus life to new students.

Industry Projects

A Machine Learning-Based Predictive Maintenance System for Early Detection of Mainshakeout Motor Failures 2026
Developed a continuous motor condition monitoring system using vibration, temperature, sound, and three-phase electrical power sensors, with data processing via ESP32 and Raspberry Pi 4 and visualization on a web dashboard.

- Integrating sensors and electrical measurement systems for motor condition monitoring.
- Developing machine learning data processing workflows and dashboards for monitoring real-time and historical data.

- Implementing automated notifications for abnormal conditions via Telegram and Gmail.
- Performing system installation, testing, commissioning, and demonstrations for the Foundry Maintenance team.

Thermoelectric Generator (TEG)-Based Power Plant Prototype 2025

Developing a power generation prototype that utilizes the temperature difference between the hot air from the Foundry Plant and the water from the cooling tower.

- Assembling a TEG circuit with a series-parallel configuration and supporting circuits for power conditioning .
- Calculating voltage, current, power, and power losses at various stages of the circuit.
- Testing and evaluating the prototype's performance and demonstrating the results to the Foundry Maintenance team.

Development and Implementation of an IoT-Based Lighting Control System 2025

Develop an IoT-based lighting control system to support remote control and monitoring of lighting at the Foundry Plant.

- Designing and assembling electrical components and control systems according to the Maintenance team's requirements.
- Performing installation, testing, and commissioning of the system on-site.
- Demonstrating the system's functions and operation to the Maintenance team.

Web-Based Spare Parts Order Status Monitoring System 2025

Developed a website to monitor the status of spare parts orders and make it easier for technicians to obtain information on order progress without the need for repeated manual confirmations.

- Develop the website's interface and functionality, including status monitoring, data management, and spare parts order history.
- Design an ordering workflow based on operational needs and information from the person in charge of spare parts management.
- Integrate Firebase Authentication and Cloud Firestore for user authentication and data management.
- Demonstrate the website to the spare parts manager and the Foundry Maintenance Team Leader.

Certifications and Achievements

- **Certificate in Supervised Machine Learning: Regression and Classification – Coursera** July 2025
- **Certificate in Cloud Fundamentals and Gen AI on AWS– Dicoding Indonesia** June 2025
- **Certificate in Fundamentals of Artificial Intelligence Implementation – Komdigi** May 2025
- **Certificate in Linear Models in Machine Learning: Fundamentals – Komdigi** May 2025
- **Certification in Getting Started with Java Programming – Dicoding Indonesia** May 2025
- **Awards for Outstanding Students in the Field of Reasoning Yogyakarta State University Year 2025** May 2025
- **Certificate Backend in Website Development – *Short Class by MySkill*** January 2025
- **Certificate Junior Network Administrator Competency, No. Reg. ICT.1272.00041 2024 – Badan Nasioanal Sertifikasi Profesi (BNSP)** November 2024
- **Finalist in the 2024 UPNVJ Electrolympic National Scientific Writing Competition** - *Team Member* November 2024
- **Award for Top Team in the 2024 National Indonesian Flying Robot Contest Racing Plane Division** - *Pilot* September 2024
- **Third Place in Region 2 of the 2024 Regional Indonesian Flying Robot Contest Division Vertical Take-Off Landing (VTOL) -** *Team Member* August 2024
- **Drone Safety and Ethics Certification Training Course (Certificate No. 8118 - 2024) Passed the Practical Test Basic Drone Maneuvers – Asosiasi Pilot Drone Indonesia** April 2024
- **Participation in the Polines Robot Competition** - *Team Leader* February 2024

Skills

- **Engineering Software:** Microsoft Office, Dev C++, Proteus, Arduino IDE, EAGLE, CX Programmer, SketchUp, FluidSIM, KiCAD, AutoCAD, VSCode.
- **Electronics and Industrial Automation:** PCB design, soldering, electronic device assembly, wiring diagrams, panel wiring, sensor and actuator integration, circuit testing and troubleshooting, and operation of mechanical and electronic tools.
- **IoT and Machine Monitoring:** Sensor integration, data acquisition, machine condition monitoring, IoT-based control systems, MQTT communication, monitoring dashboards, and automated notifications.
- **Programming and Web Development:** C/C++, Python, JavaScript, HTML, CSS, Next.js, Tailwind CSS, MySQL, MongoDB, Firebase.
- **Interpersonal Skills:** Teamwork, leadership, adaptability, initiative, and time management.
- **Languages:** Indonesian (Fluent), English (Intermediate).